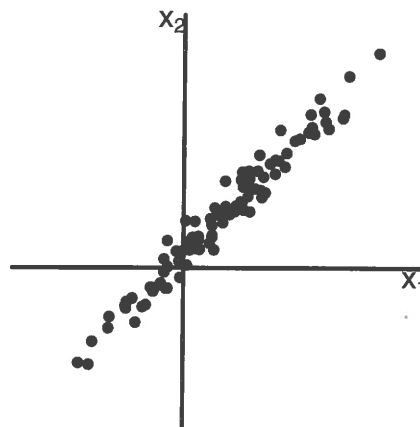


1 Principal Components Analysis

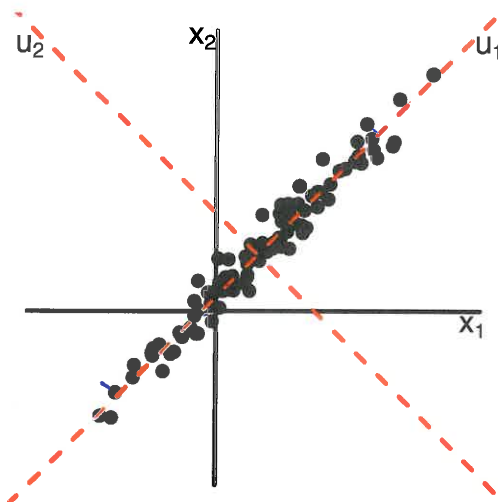
- 2 A standard, classic tool for dimension reduction is **principal components**
3 **analysis (PCA)**. It has some drawbacks, mostly in that it is best suited to
4 find **linear** structure in data, but in the right situation it can be a very effective tool. First, consider the synthetic data shown in the following figure.



- 1 **Exercise:** Comment on the structure seen in the pair of variables shown in
2 the previous figure. Would you say that the current pair of axes is the most
3 “natural” way to represent the data?

4 Clearly a strong linear relationship. It would be
5 more natural to represent these data
6 using an axis system that corresponds
7 to this relationship

- 1 The figure below shows the same data, with a new (u_1, u_2) coordinate system shown as the dashed lines.



3

- 4 These axes seem to be a more natural representation of the data. In particular, we note that the u_1 axis captures the major source of variability in the data. These axes are formed from the **principal components** of the data.

1 Formal Setup of PCA

- 2 Suppose we have observed n vectors $\mathbf{x}_i$, each of which is of dimension p :

$$\mathbf{x}_i = (x_{i1}, x_{i2}, \dots, x_{ip})^T, \quad i = 1, 2, \dots, n.$$

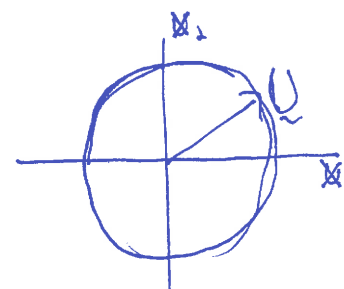
- 4 For example, $\mathbf{x}_i$ could be a single yield curve, with $p \approx 10$.

- 5 Consider the collection of all possible linear combinations of the original variables formed by different choices of $\mathbf{u}$:

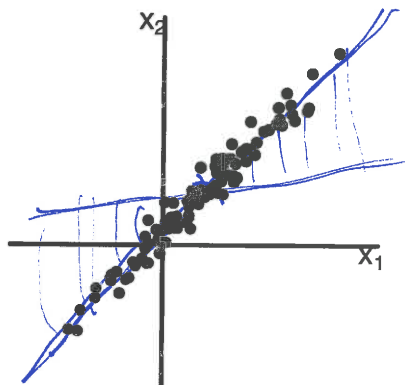
$$v_i = \sum_{j=1}^p u_j (x_{ij} - \bar{x}_j), \quad i = 1, 2, \dots, n$$

- 8 where $\bar{x}_j$ is the mean of the j^{th} variable, subject to

$$\sum_{j=1}^p u_j^2 = 1.$$



- 1 Hence, we can think of each choice of $\mathbf{u}$ as being a different **direction** cen-
- 2 tered on the sample mean.
- 3 The new measurements $v_1, v_2, \dots, v_n$ are the **projections** of the original ob-
- 4 servations onto this new direction.
- 5 **We now ask:** What choice of $\mathbf{u}$ maximizes the sample variance of the re-
- 6 sulting numbers $v_1, v_2, \dots, v_n$?



1 The Math:

- 2 If $\mathbf{A}$ is a symmetric, positive definite matrix then the choice of $\mathbf{u}$ that max-
- 3 imizes $\mathbf{u}^T \mathbf{A} \mathbf{u}$ subject to $\mathbf{u}^T \mathbf{u} = 1$ is the first eigenvector of $\mathbf{A}$.

- 4 Recall that

$$V(\mathbf{u}^T \mathbf{x}) = \mathbf{u}^T \Sigma \mathbf{u}$$

$$\mathbf{u}^T \mathbf{u} = \sum_{i=1}^p u_i^2$$

- 6 where $V(\mathbf{x}) = \Sigma$.

- 7 We approximate Σ using the sample covariance matrix $\hat{\Sigma}$.

- 8 The leading eigenvector of $\hat{\Sigma}$ is the first principal component.

- 9 The additional principal components $\mathbf{u}_2, \mathbf{u}_3, \dots, \mathbf{u}_p$ are found by finding
- 10 successive eigenvectors of $\hat{\Sigma}$.

1 **Some comments:**

- 2 A new coordinate system is being constructed in which the p -dimensional
3 data are represented. The center of this coordinate system is the sample
4 mean of the data, but the axes are the principal components $u_1, u_2, \dots, u_p$.
- 5 When expressed in this new coordinate system, each of the dimensions
6 have sample correlation of zero.
- 7 The amount of “variance explained” by each principal component is less
8 than each of the previous principal components.

- 1 In many situations, we will choose some cutoff $q \ll p$ such that we will
2 only retain the first q axes in our new coordinate system. This will be
3 because these first q components “capture” most of the variability in the
4 data.
- 5 For instance, in the toy example above, most of the variability in the data
6 is “captured” by the first axis u_1 .
- 7 In typical applications p will be very large, and a low-dimensional repre-
8 sentation will be of great value.

1 Back to the Yield Curves

2 Our previous inspection of some yield curves suggested that these curves
3 share a common shape, and that it does not require ten numbers to de-
4 scribe changes in this shape from day to day.

5 This is an ideal situation to consider dimension reduction. The simplicity
6 of the shapes of the curves suggests that PCA is worth trying.

7 For this analysis we will use the yield curves from 2010 to the present. We
8 need to extract the relevant rates to run the PCA. Note that there are 11
9 rates over this time period.

```
> YCrates = YCdata[YCdata$Date > "2010-01-01", -1]
```

← remove
Date column

1 There is one row that is filled with zeros:

```
> YCrates = YCrates[-which(apply(YCrates, 1, sum) == 0),]
```

2 We want to characterize shifts in the yield curves, so we transform into the
3 day-to-day change in the rates:

```
> YCshifts = apply(YCrates, 2, diff)
```

4 The function `princomp()` runs the PCA. The `subset` argument is used
5 to skip over the rows with NA.

```
> pcaout = princomp(YCshifts,  
+ subset=complete.cases(YCshifts))
```

- 1 The following output is useful:

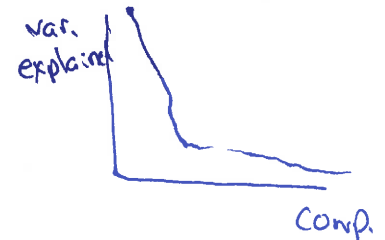
> **summary**(pcaout)

Importance of components:

	Comp.1	Comp.2	Comp.3	Comp.4
Standard deviation	0.1132046	0.03442868	0.02476141	0.01669611
Proportion of Variance	0.8253307	0.07633800	0.03948671	0.01795273
Cumulative Proportion	0.8253307	0.90166871	0.94115543	0.95910815

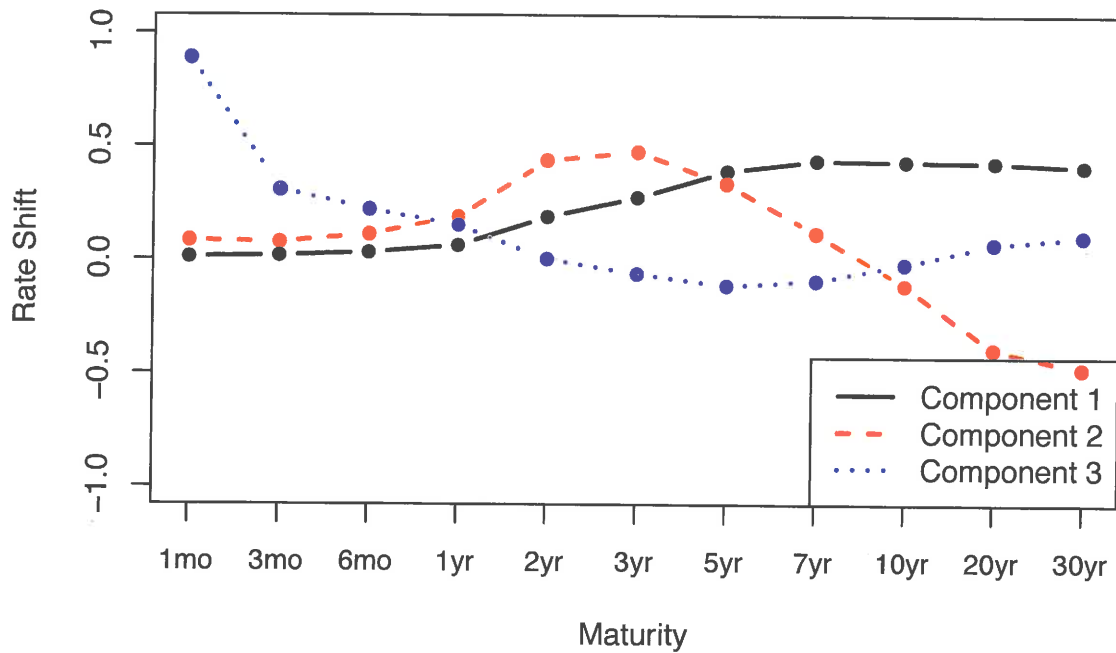
	Comp.5	Comp.6	Comp.7	Comp.8
Standard deviation	0.01431992	0.011679546	0.009897267	0.009073786
Proportion of Variance	0.01320631	0.008785207	0.006308566	0.005302457
Cumulative Proportion	0.97231446	0.981099670	0.987408236	0.992710693

	Comp.9	Comp.10	Comp.11
Standard deviation	0.006819678	0.005906935	0.005637764
Proportion of Variance	0.002995214	0.002247111	0.002046982
Cumulative Proportion	0.995705907	0.997953018	1.000000000



- 2 This shows the proportion of variance explained by each component.

- 1 The first three components account for approximately 95% of the variability
- 2 in the shifts in the yield curves.
- 3 The actual components (directions) are called the **loadings**. Look at
- 4 `pcaout$loadings` to see these vectors. The following plot shows the
- 5 first three components/loadings.
- 6 The interpretation is as follows: The day-to-day change in the yield curves
- 7 can mostly be modelled as a linear combination of the three curves shown
- 8 in this figure. The weight placed on each component (called the **score**) will
- 9 change from day to day.



1

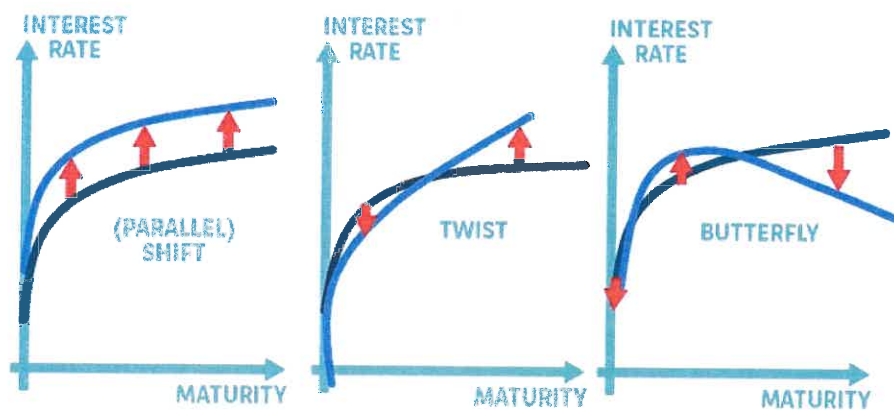


Figure 1: Figure from advisoranalyst.com

- 1 The scores give a **representation** of the shifts in yield curves. In fact, this is
- 2 related to a common way to characterize how the yield curve has changed.
- 3 The first three components are commonly given the names "parallel shift,"
- 4 "twist," and "butterfly."

- 1 The `predict()` function, when applied to the output of `princomp()`
- 2 will return the position of yield curve shift in this new representation. Note
- 3 that R is a little picky on the format. The new shift must come in as a data
- 4 frame with column names the same as used in the data.

```
> newshift = matrix(c(0.00, 0.00, 0.01, 0.01, 0.01, 0.07, 0.05, 0.05,
+                     0.03, 0.03, 0.04), nrow = 1)
> newshift = data.frame(newshift)
> names(newshift) = colnames(YCshifts)
> predict(pcaout, newshift)
```

	Comp.1	Comp.2	Comp.3	Comp.4	Comp.5	Comp.6
[1,]	0.1060112	0.02753681	-0.005937651	0.003564653	-0.00352043	0.003759287

	Comp.7	Comp.8	Comp.9	Comp.10	Comp.11
[1,]	-0.005728047	-0.03691347	-0.0005428808	0.009590538	0.006370914

1 Scaling Variables for PCA

- 2 In cases where the variables being incorporated into PCA are on different
- 3 scales, it is crucial that the variables be standardized prior to the analysis.
- 4 It is customary to scale variables so that each has sample mean of zero, and
- 5 sample variance of one.
- 6 This is equivalent to finding the eigenvectors of the correlation matrix in-
- 7 stead of the covariance matrix. In fact, with function `princomp()`, one
- 8 can simply use argument `cor = T` to achieve this.

1 Nonlinear Dimension Reduction

2 Consider the figure on the following slide. In this case, an individual
3 “observation” is an image of a face. These faces are mapped into a two-
4 dimensional space. The faces along the bottom show how the faces evolve
5 along the line in the upper right of the figure.

6 This illustrates the use of **nonlinear dimension reduction** on high-
7 dimensional data. PCA would not have been a good choice in this
8 case because individual images cannot be approximated by a linear
9 combination of a “basis” faces.

1 **Exercise:** What is the utility of the representation shown in the previous
2 figure?

3 _____

4 _____

5 _____

6 _____

7 _____

8 _____

9 _____

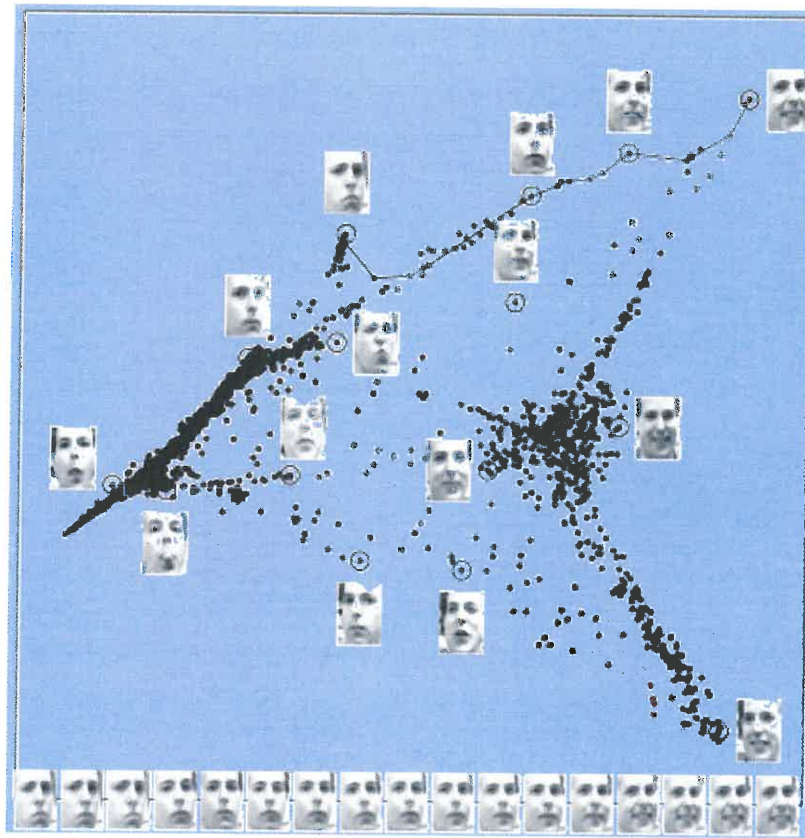
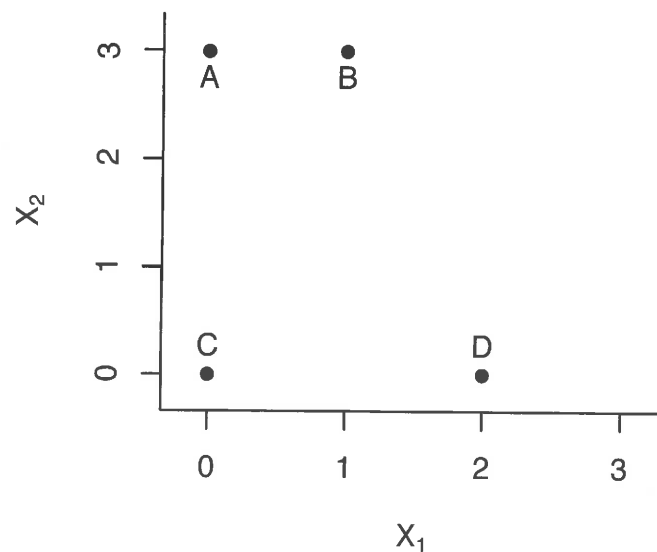


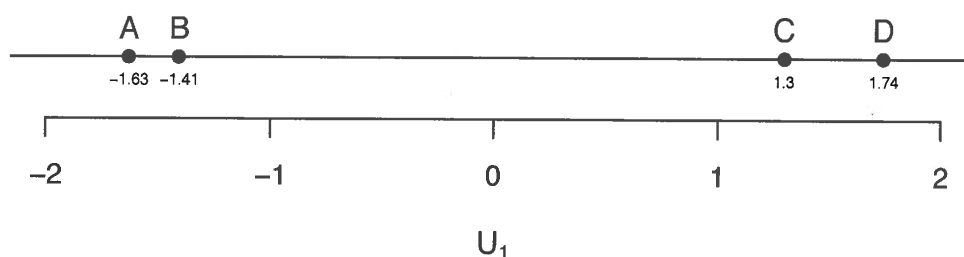
Figure 2: From Roweis and Saul (2000), *Science*

1 Multidimensional Scaling

- 2 Consider a synthetic case in which we have four points in two-dimensions,
- 3 labelled as in the following figure.



- 1 Imagine calculating all of the pairwise distances between the four points
- 2 shown in the above figure.
- 3 Then ask: **What one-dimensional representation of the data preserves**
- 4 **those distances to the greatest extent possible?**
- 5 The answer is addressed by **multidimensional scaling (MDS)**.
- 6 The result, using the classic approach, is shown in the figure below.



- 1 **Exercise:** What is the practical motivation behind trying to find a low-
- 2 dimensional representation that preserves distances?

3 We've already discussed why we want a low-

4 dimensional rep.

5 Observations which are "close" in high-dimensions

6 are believed to be "similar", i.e. distance

7 is a proxy for similarity

8

9

10 Finding low dim. rep. that preserves this measure of similarity (to the extent possible)

1 In R, classic multidimensional scaling is implemented through function
2 `cmdscale()`.

3 Note in the example below how the `dist()` function is used in conjunc-
4 tion with `cmdscale()`. The argument `k` controls how many dimensions
5 should be returned.

```
> mdsout = cmdscale(  
+   dist(YCshifts[complete.cases(YCshifts),]), k=2)
```

6 In this example, each “observation” is a single shift in the yield curve. We
7 calculate the distances between all pairs of yield curves, and then seek
8 the two-dimensional mapping that preserves these distances to the extent
9 possible.

1 **Exercise:** Compare the results from multidimensional scaling to those
2 found using PCA. What conclusion can you draw?

3 PCA attempts to find the lower dim. rep.
4 that preserves the original Euclidean
5 distance, to the extent possible